
Exercises

Advanced Lesson 1

Your Name: Mikey Ferguson

Your Math Class: MATH 7013 – Advanced Mathematical Modeling

1: Inside this notebook:

- Place the Tool Bar at the top of this notebook.

menu: Window $\Rightarrow$ Show Toolbar

Check out the "fast" access to different cell styles located at the left edge of the Toolbar.

Open *Mathematica*'s internal [Help Documentation Center](#).

- Change the "Magnification" (located at the lower right-hand corner) from the default 100% to a more readable size (like 125% or 150% etc.).
- Search for the *Mathematica* function **Integrate** using [Help Documentation Center](#):

Type: **Integrate** $\Rightarrow$ and follow the link to **Integrate** ($\int$) and evaluate the **Basic Examples** in the [Help Documentation Center](#).

- Click on each of the following 7 blue links to the [Help Documentation Center](#).
Read the contents of each tutorial and also **evaluate** each of the tutorial Input cells:

Getting Started

Your First *Mathematica* Calculations

Arithmetic

Defining Variables

Defining Functions

Some Mathematical Functions

Symbolic Calculations

Numerical Calculations

- Don't worry about an answer--just do it!

2: Use *Mathematica*'s internal [Help Documentation Center](#) to locate information on the "numerical" function **N[*expr*]**:

Type: **n** $\Rightarrow$ Click to search and locate $\Rightarrow$ **N[*expr*]**

Note the two ways you can use **N**:

N[*expr*]

gives the numerical value of *expr*.

N[*expr*, *n*]

attempts to give a result with *n*-digit precision.

- Modify the "Basic Examples" in [Help Documentation Center](#) to calculate: **N[1/13]**
- Modify the "Basic Examples" in [Help Documentation Center](#) to calculate: **N[1/13 , 50]**
- Cut and Paste the two Input-Output cell pairs containing the above results into this notebook just below this text cell.

N[$\frac{1}{13}$]

N[$\frac{1}{13}$, 50]

3: Insert your answers to the following questions by inserting new Input Cells for each question immediately below this Text Cell.

Enter the *Mathematica* code which accomplishes the following:

- **Clear[a, A];**
Evaluate the product: **a*A**
(Use one Input cell for both lines and be sure to use the ";")
- Check the values assigned to both: **a** and: **A** using "?"
(Use two Input cells.)
- Assign **5** to **a**

- Evaluate **a*A**
(Use a new Input cell.)
- Check the values assigned to both: **a** and: **A** using "?"
(Use two Input cells.)
- Assign **6** to **A**
- Evaluate **a*A** again
(Use a new Input cell.)
- Check the values assigned to both: **a** and: **A** using "?"
(Use two Input cells.)
- Assign **7** to **A**
- Evaluate **a*A**
- Check the values assigned to both: **a** and: **A** using "?"
- Briefly explain your results (in a Text Cell located below your Input-Output cells):

```
Clear[a, A];
```

```
a A
```

```
?a
```

```
?A
```

```
a = 5
```

```
a A
```

```
?a
```

```
?A
```

```
A = 6
```

```
a A
```

```
?a
```

```
?A
```

```
A = 7
```

```
a A
```

?a

?A

Mathematica is case sensitive, so a and A are two different symbols. Before any assignment neither has a value and a A stays as a product. After $a = 5$ the product becomes $5 A$; after $A = 6$ it becomes 30 ; after $A = 7$ it becomes 35 . Each ? shows the value that symbol is holding at that moment.

4: Insert your answers to the following questions by inserting new Input Cells immediately below this Text Cell.

- Check the meaning assigned to: **Sin** using "?"
(Then follow the link >> to [Help Documentation Center](#) to see what is there.)
- Check the meaning assigned to: **sin** using "?"
- Explain the results (in a Text Cell).

?Sin

?sin

Sin is the built-in sine function, so ?Sin shows its usage message and a documentation link. sin with a small s is not a Mathematica name at all, just an undefined symbol, so ?sin shows no value. Every built-in name begins with a capital letter.

5: Insert your answers to the following questions by inserting new Input Cells immediately below this Text Cell.

Enter the *Mathematica* code which accomplishes the following:

- **Clear[a, A, feet, sec];**
Evaluate **a/A**
(Use one Input cell for both lines and be sure to use the ";")
- Check the values assigned to both: **a** and: **A**
- Assign: **5 feet / sec** to: **A**
(i.e. **A = 5 feet/sec**)
- Evaluate **a/A**
- Check the values assigned to both: **a** and: **A**
- Assign: **10 feet** to: **a**
- Evaluate **a/A**
- Check the values assigned to both: **a** and: **A**
- Check the values assigned to both: **feet** and: **sec**

- Explain the results (in a Text Cell).

```
Clear[a, A, feet, sec];
```

```
a
—
A
```

```
?a
```

```
?A
```

```
A =  $\frac{5 \text{ feet}}{\text{sec}}$ 
```

```
a
—
A
```

```
?a
```

```
?A
```

```
a = 10 feet
```

```
a
—
A
```

```
?a
```

```
?A
```

```
?feet
```

```
?sec
```

feet and sec are never given values, so they stay in the answer as ordinary symbols. That is why a/A comes out as 2 sec: the feet cancel and $10/5 = 2$ is left. ?feet and ?sec show no value, and that is exactly what lets them behave like units.

6: Insert your answers to the following questions by inserting new Input Cells immediately below this Text Cell.

Enter the *Mathematica* code which accomplishes the following:

- **Clear[x, f]**
Define a function: **f** of the variable: **x** to be the **Sin[x]**.
(Remember to use the "_".)
(Use one Input cell for both lines of code and be sure to use the ";")
- Evaluate the Sine of π and $\pi/2$ and **6.0** using your function: **f** and by using the Basic Math Input Palette button  to enter π .
(i.e. evaluate **f[π]**, **f[$\pi/2$]**, **f[6.0]**)
- Explain the results (in a Text Cell).

- Check the value assigned to the function: **f** using "?"
- Evaluate the derivative of the function: **f** with respect to **x**, i.e. $\partial_x f[x]$, by using the Basic Math Input Palette function .
(The Output cell will contain: **Cos[x]**)
- Evaluate the derivative of the function: **f** with respect to **x** using the alternate notation **f'[x]**
(The Output cell will contain: **Cos[x]**)
- Evaluate the integral of the function: **f** with respect to **x**, i.e. $\int f[x] dx$, by using the Basic Math Input Palette function .
- Solve the equation: **3 f[x] + 4 == 0** for **x** using the **Solve[...]** command. Remember, the **==** is typed as **=** followed by another **=**.
- Plot **f[x]** for **0 ≤ x ≤ 2 π** using the **Plot[...]** command and by using the Basic Math Input Palette button  to enter π .

```
Clear[x, f];
f[x_] = Sin[x];
```

```
{f[Pi], f[Pi/2], f[6.]}
```

```
?f
```

```
 $\partial_x f[x]$ 
```

```
f'[x]
```

```
 $\int f[x] dx$ 
```

```
Solve[3 f[x] + 4 == 0, x]
```

```
Plot[f[x], {x, 0, 2 Pi}]
```

$f[\text{Pi}]$ is exactly 0 and $f[\text{Pi}/2]$ is exactly 1, because Pi is the exact symbol and Mathematica knows those values. $f[6.0]$ is -0.279415, an approximate number, because 6.0 is approximate: one approximate input makes the whole answer approximate.

7: Insert your answers to the following questions by inserting new Input Cells immediately below this Text Cell.

Enter the *Mathematica* code which accomplishes the following:

- **Clear[a, x, f, y]**
- Define a function: **f** of the specific symbol: **x** and only **x** to be x^2 by **f[x] = x²**. Use the Basic Math Input Palette function .
(Note that there is no "_" used in the above definition.)

- Without clearing the definition of **f** define a function: **f** of the variable: **x** to be e^x by **f[x_] = e^x**
(Note that "_" is used in the above definition.)
Use the Basic Math Input Palette functions  and  to get e^x and not e^x .
- Without clearing the definition of **f** define a function: **f** at **x = 2** to be π by **f[2] = π**
Use the Basic Math Input Palette button .
- Evaluate: **f[x]**
- Evaluate: **f[a]**
- Evaluate: **f[y]**
- Evaluate: **f[2]**
- Evaluate: **f[3]**
- Explain the results (in a Text Cell):
- Check the values assigned to the function: **f** using "?"
- Explain the results (in a Text Cell):

```
Clear[a, x, f, y];
```

```
f[x] = x^2;
```

```
f[x_] =  $e^x$ ;
```

```
f[2] =  $\pi$ ;
```

```
{f[x], f[a], f[y], f[2], f[3]}
```

```
?f
```

Three definitions are stored and none of them replaces the others. $f[x] = x^2$ applies only to the literal symbol x , $f[2] = \pi$ only to the literal 2, and $f[x_] = E^x$ to any argument. Mathematica uses the most specific definition that fits, so $f[x]$ gives x^2 and $f[2]$ gives π , while $f[a]$, $f[y]$ and $f[3]$ fall through to E^a , E^y and E^3 . `?f` lists all three.

8: Insert your answers to the following questions by inserting new Input Cells immediately below this Text Cell.

The following exercise illustrates a circumstance where using delayed assignment "**:=**" and immediate assignment "**=**" do NOT give the same result. The idea is to define a function which will generate random real numbers between 0 and 1 scaled by the size of x .

When immediate assignment "**=**" is used to define the function **f1[x_]**

```
f1[x_] = RandomReal[] * x
```

then the right hand side will be evaluated immediately once and only once. Therefore
RandomReal[]

will be called once and only once and therefore would have only one value no matter how many other times you call the function **f1**.

When delayed assignment `":="` is used to define the function **f2[x_]**

f2[x_] := RandomReal[]*x

then the right hand side will NOT be evaluated until the function **f2** is called later. And the function

may be called many many different times (as would occur when you generate a plot of the function **f2**).

Therefore each and every time the function **f2** is called, then the right-hand side will be evaluated

anew and thereby generating a new and different evaluation of **RandomReal[]**.

- Enter the *Mathematica* code which accomplishes the following:

Clear **f1,f2** and **x**

Find information about **RandomReal** using `"?"`

Define **f1[x_]** to be **RandomReal[]*x** using immediate assignment `"="`

Evaluate **f1[x]**

Evaluate **f1[x]**

Evaluate **f1[x]**

Plot **f1** for $0 \leq x \leq 1$

Find information about **f1** using the `"?"` command

Define **f2[x_]** to be **RandomReal[]*x** using delayed assignment `":="`

Evaluate **f2[x]**

Evaluate **f2[x]**

Evaluate **f2[x]**

Plot **f2** for $0 \leq x \leq 1$

Find information about **f2** using the `"?"` command

- Explain the above results (in a Text Cell):

```
Clear[f1, f2, x];
```

```
?RandomReal
```

```
f1[x_] = RandomReal[] x;
```

```
{f1[x], f1[x], f1[x]}
```

```
Plot[f1[x], {x, 0, 1}]
```

```
?f1
```

```
f2[x_] := RandomReal[] x;  
{f2[x], f2[x], f2[x]}
```

```
Plot[f2[x], {x, 0, 1}]
```

```
?f2
```

With = the right-hand side is worked out once, when the definition is made, so a single random number is chosen and frozen: the three values of f1[x] are identical and the plot is a straight line. With := nothing is worked out until the function is called, so a new random number is drawn every time: the three values differ and the plot is a band of scattered points, because Plot calls the function once per plotted point.

9: Insert your answers to the following questions by inserting new Input Cells immediately below this Text Cell.

- Clear **x** and **n**.
- Evaluate the derivative of the function: x^n with respect to **x** using the Basic Math Input Palette functions  and .
- Evaluate the integral of the function: x^n with respect to **x**, using the Basic Math Input Palette function .
- Solve the algebraic equation: $x + \sqrt{x} == 9$ for **x** using the **Solve[...]** command and the Basic Math Input Palette function .
Remember, the == is typed as = followed by another =.
- Plot $x^3 - 2x + 1$ for $-2 \leq x \leq 2$ using the **Plot[...]** command and by using the Basic Math Input Palette button  to enter x^3 .
- Explain the above results (in a Text Cell):

```
Clear[x, n];
```

```
 $\partial_x x^n$ 
```

```
 $\int x^n dx$ 
```

```
Solve[x +  $\sqrt{x}$  == 9, x]
```

```
Plot[x3 - 2 x + 1, {x, -2, 2}]
```

The two calculus results are the ordinary power rules, stated for a symbolic n. Solve returns only $x = (19 - \text{Sqrt}[37])/2$, because the other root of the underlying quadratic would require a negative square root and does not satisfy the original equation. The plot of $x^3 - 2x + 1$ rises to a local maximum near $x = -0.8$, dips to a local minimum near $x = 0.8$, and rises again.

10: Insert your answers to the following questions by inserting new Input Cells immediately below this Text Cell.

- Using the "?" command, find information about: **Style**
(Then follow the link >> to [Help Documentation Center](#) to glimpse how **Style** works.)

Using the "?" command and the wild card " * ", determine:

- All *Mathematica* commands that start with: **Style**
(there are about 12).
- All *Mathematica* commands that end in: **Style**
(there are about 45).
- All *Mathematica* commands that contain the word: **Style**
(there are about 65).

?Style

Names["Style*"]

Length[Names["Style*"]]

Names["*Style"]

Length[Names["*Style"]]

Names["*Style*"]

Length[Names["*Style*"]]